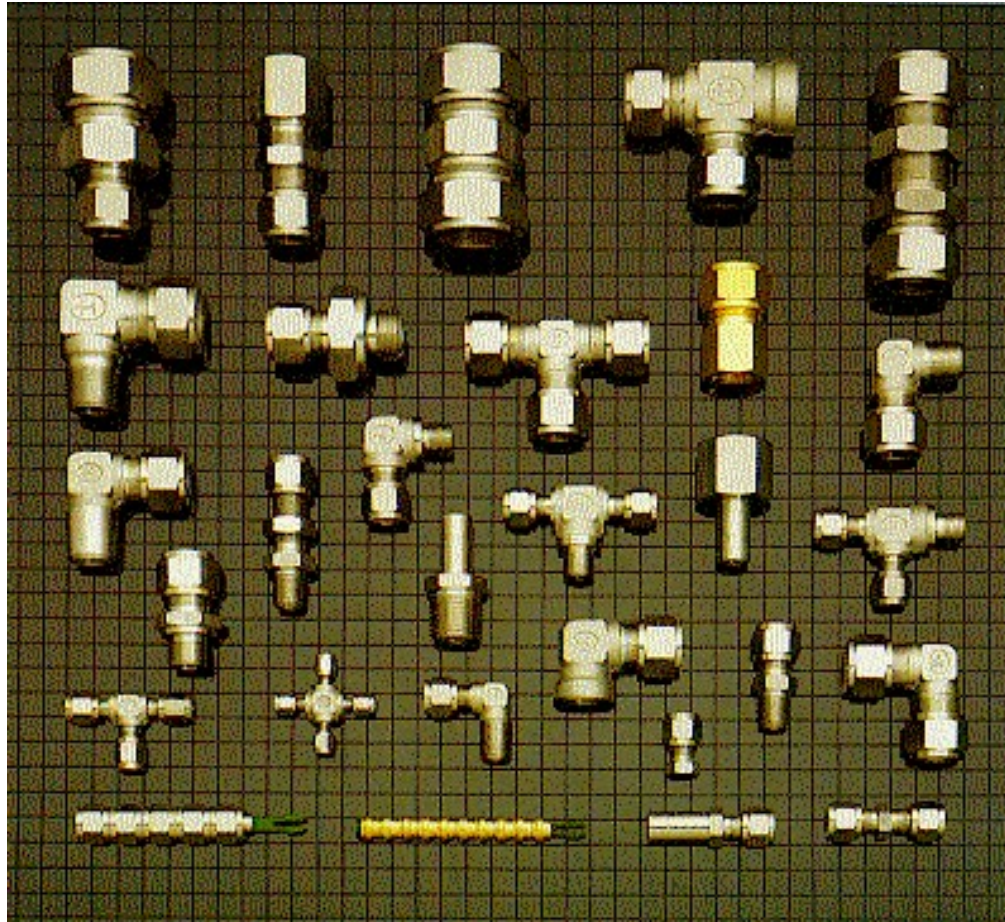


Lecture 08: Parameter Estimation, χ^2 Minimization



A bunch of fittings

Physics 4730
Fall Semester 2026

Undersampling Bias in the Real World!

PRL **104**, 161101 (2010)

PHYSICAL REVIEW LETTERS

week ending
23 APRIL 2010

Indications of Proton-Dominated Cosmic-Ray Composition above 1.6 EeV

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(The High Resolution Fly's Eye Collaboration)

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We report studies of ultrahigh-energy cosmic-ray composition via analysis of depth of air shower maximum ($X_{\max}$), for air shower events collected by the High-Resolution Fly's Eye (HiRes) observatory. The HiRes data are consistent with a constant elongation rate $d(X_{\max})/d(\log E)$ of $47.9 \pm 6.0(\text{stat}) \pm 3.2(\text{syst})$ g/cm²/decade for energies between 1.6 and 63 EeV, and are consistent with a predominantly protonic composition of cosmic rays when interpreted via the QGSJET01 and QGSJET-II high-energy hadronic interaction models. These measurements constrain models in which the galactic-to-extragalactic transition is the cause of the energy spectrum ankle at 4×10^{18} eV.

DOI: 10.1103/PhysRevLett.104.161101

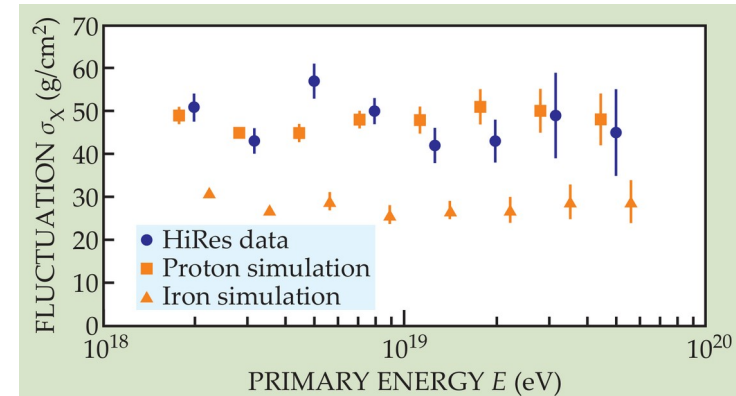
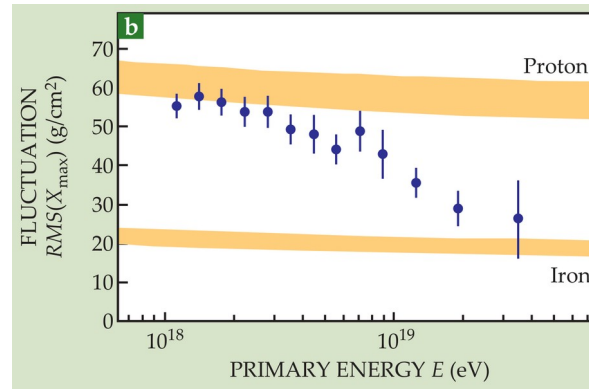
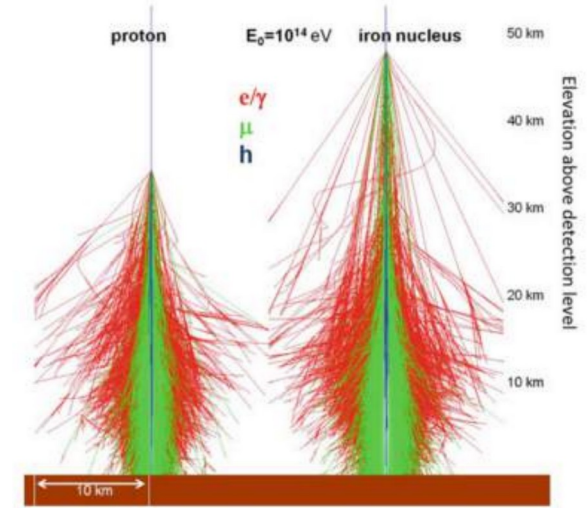
PACS numbers: 98.70.Sa, 95.85.Ry, 96.50.sh, 96.50.sd

The observation of a break in the cosmic-ray energy spectrum at approximately 6×10^{19} eV [1–3] provides evidence that the highest energy cosmic rays are both extragalactic and protonic [4,5]. Direct evidence for a proton-dominated composition from air shower data would lend further support to this model, as would the observation of a transition from (heavy) galactic to (light) extragalactic cosmic rays at lower energies. A second feature, the ankle of the energy spectrum at 4×10^{18} eV, may be indicative of this transition or it may further strengthen the model in which the end of the cosmic-ray spectrum is shaped by interactions with the cosmic microwave background [6]. Composition studies can provide decisive evidence in the choice between interpretations.

An important clue to chemical composition which is accessible to air fluorescence observatories is the depth of shower maximum $X_{\max}$ of cosmic ray induced extensive air showers. Simple arguments [7,8] show that the average value of air shower maximum $\langle X_{\max} \rangle$ will depend logarithmically on the primary energy and atomic mass, and that the elongation rate $d(X_{\max})/d \log E$ will be constant for

unchanging primary compositions. Further, to first order, a nucleus-induced shower may be thought of as a superposition of showers induced by single nucleons. Therefore, due to averaging effects, we also expect the width of the $X_{\max}$ distribution at a given energy to be sensitive to the atomic mass of the primary.

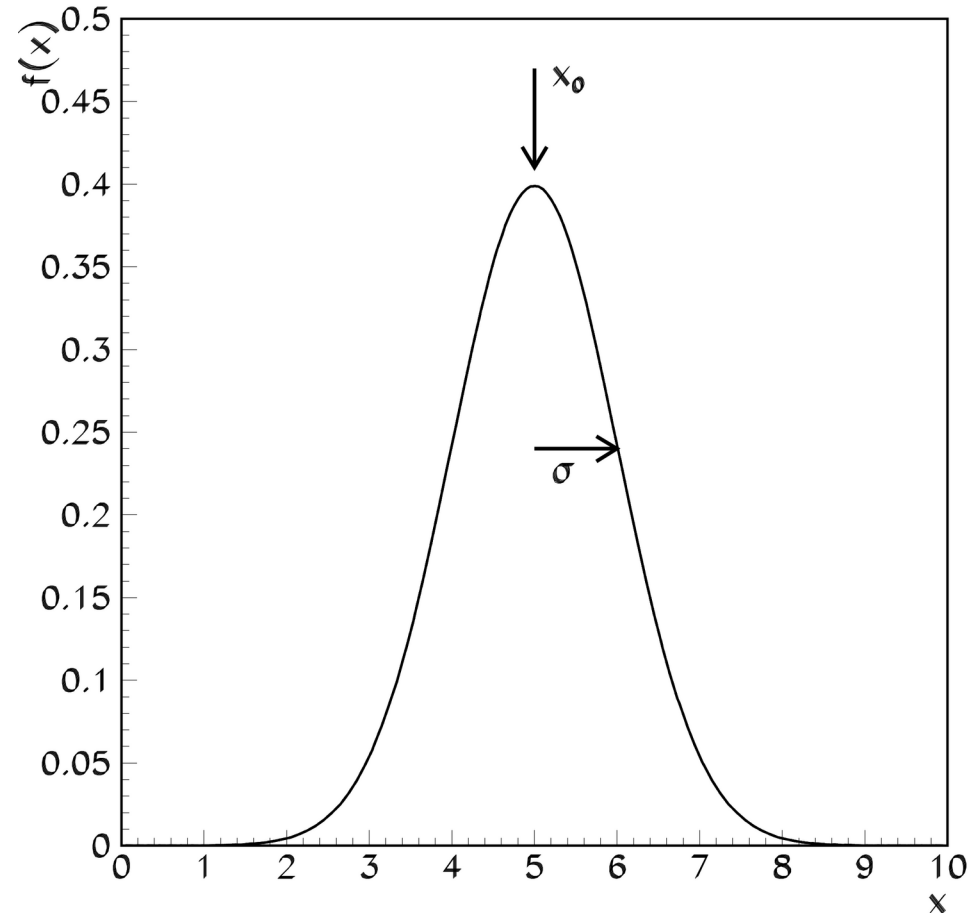
The two fluorescence observatories of the High-Resolution Fly's Eye collected data in stereoscopic mode between December 1999 and April 2006. Located on the U.S. Army Dugway Proving Ground in Utah, at a mean elevation of 1575 m MSL, a mean latitude of 40.16° N, and separated by 12.6 km, the observatories operated on clear moonless nights. Each detector consisted of an array of telescope modules, each module included a mirror of 3.7 m² effective area which focused UV light from air showers on a 16×16 photomultiplier tube (PMT) camera. The field of view of each PMT subtended a 1° diameter cone of the sky. The HiRes-I detector covered nearly 360° in azimuth, 3°–17° in elevation and was read out by means of sample-and-hold electronics, while the HiRes-II detector covered 3°–31° in elevation and was read out by a



Fitting Data with Gaussian-Distributed Errors

Least-Squares Fitting

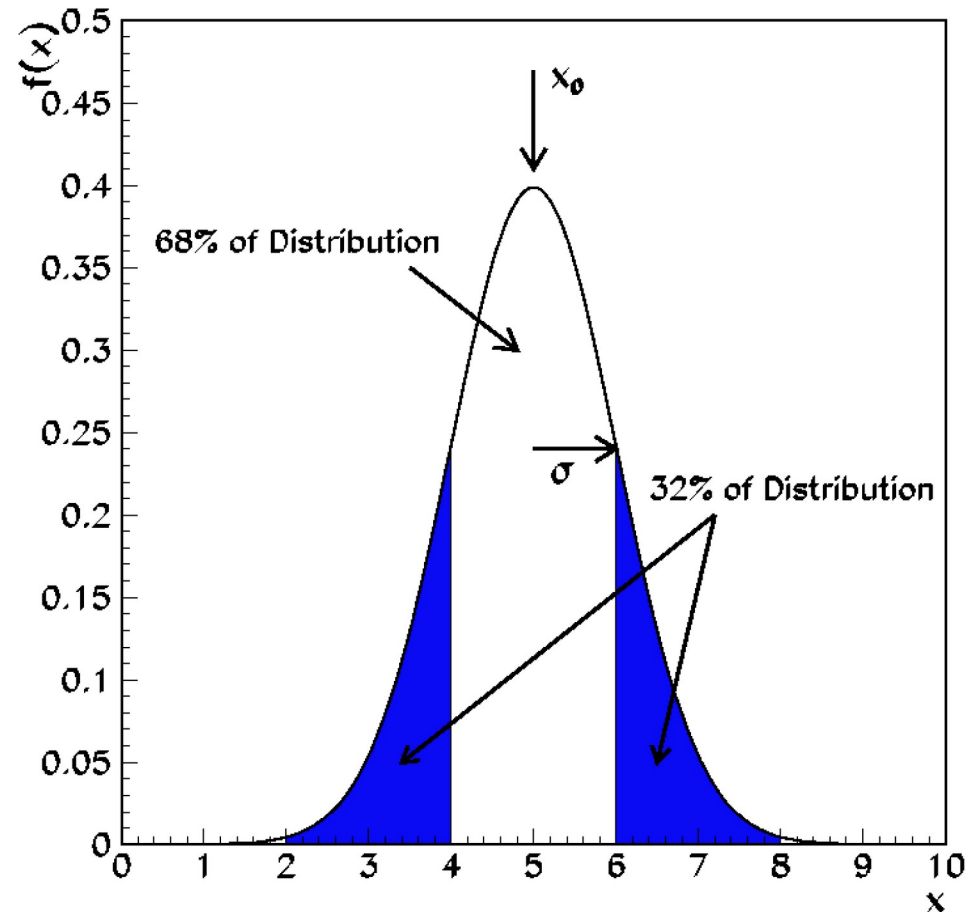
- Simplest case: straight line fit to data with *normally distributed (Gaussian)* uncertainties.
- What does this mean?
 - Measure quantity x , N times.
 - As $N \rightarrow$ large, parent probability distribution $\rightarrow$ Gaussian



$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\bar{x})^2/2\sigma^2}$$

Gaussian Errors

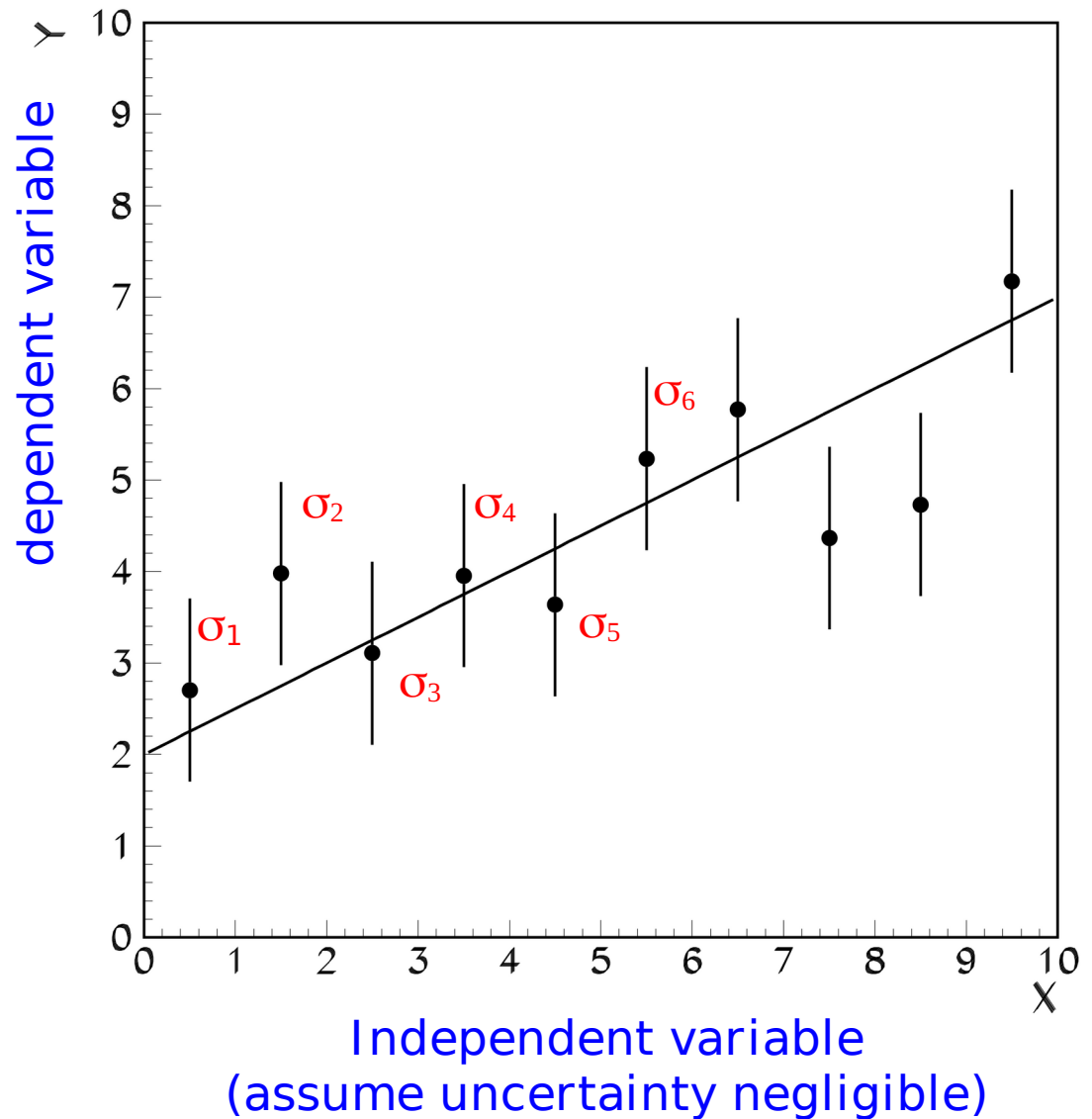
- Note that one sigma (1σ) error interval is *not* the maximum possible excursion from X_0 :
 - 32% > 1σ
 - 4.5% > 2σ
 - 0.27% > 3σ



$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\bar{x})^2/2\sigma^2}$$

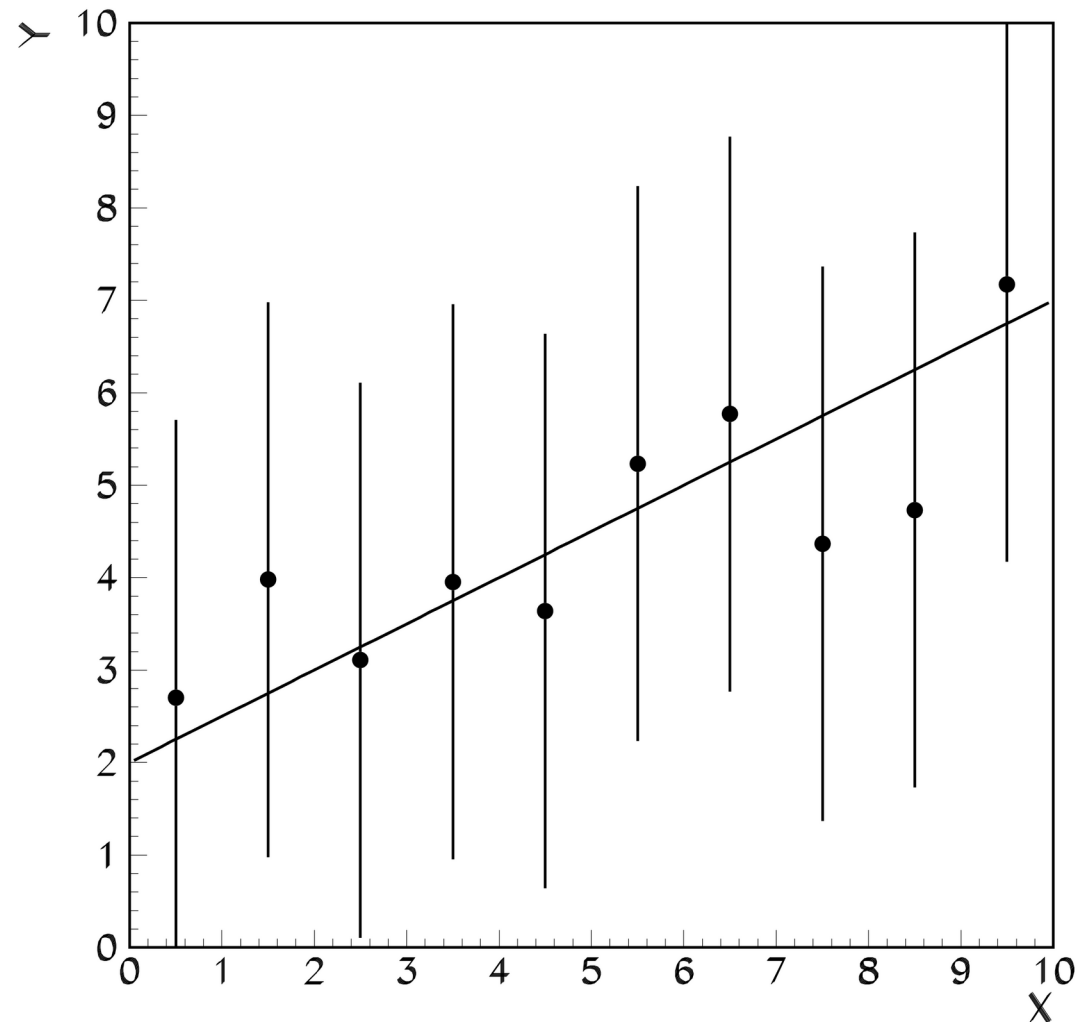
Use “2/3rds Rule” as a Rough Guide

- Spread of points, error bars consistent with normal error distribution.



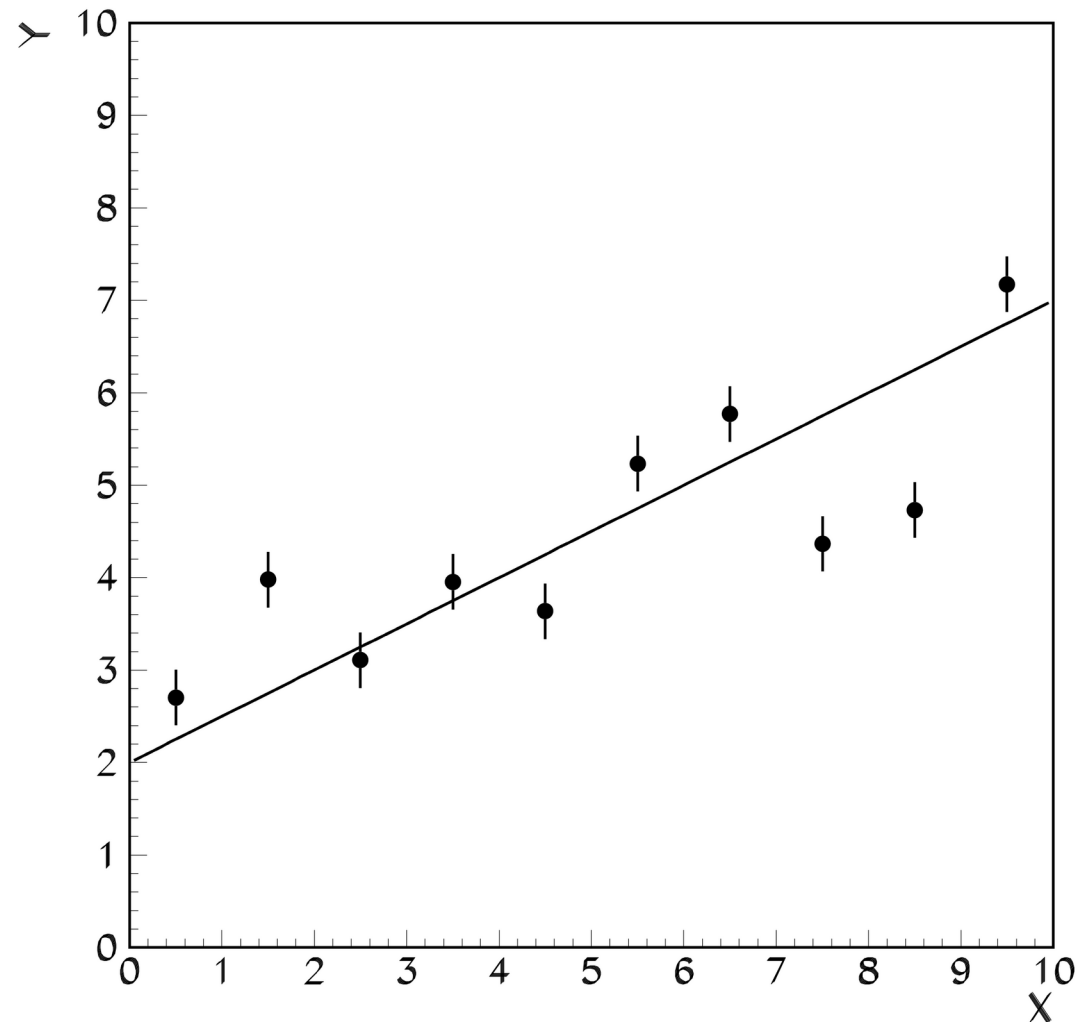
Use “2/3rds Rule” as a Rough Guide

- What is the problem here?



Use “2/3rds Rule” as a Rough Guide

- What is the problem here?



Problem of Straight-Line Fitting

- Remaining problem: I have a set of points

$$(x_i, y_i \pm \sigma_i)$$

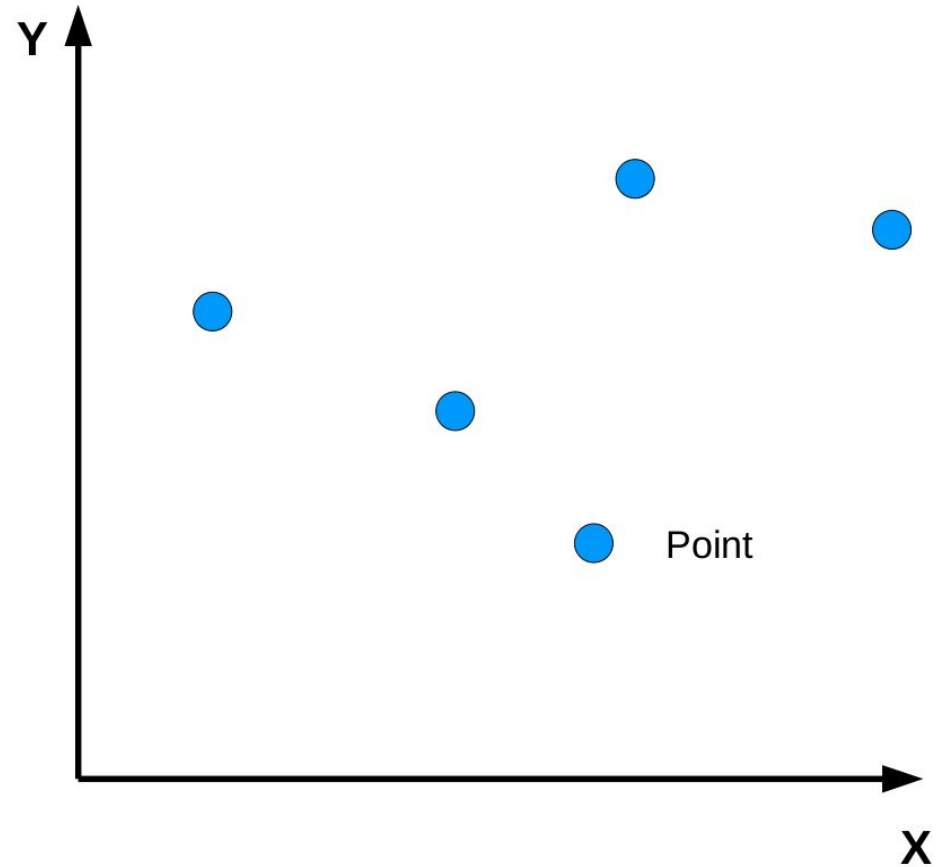
which I expect to follow a linear relationship

$$y = A + Bx$$

What is my best estimate for the coefficients A , B ?

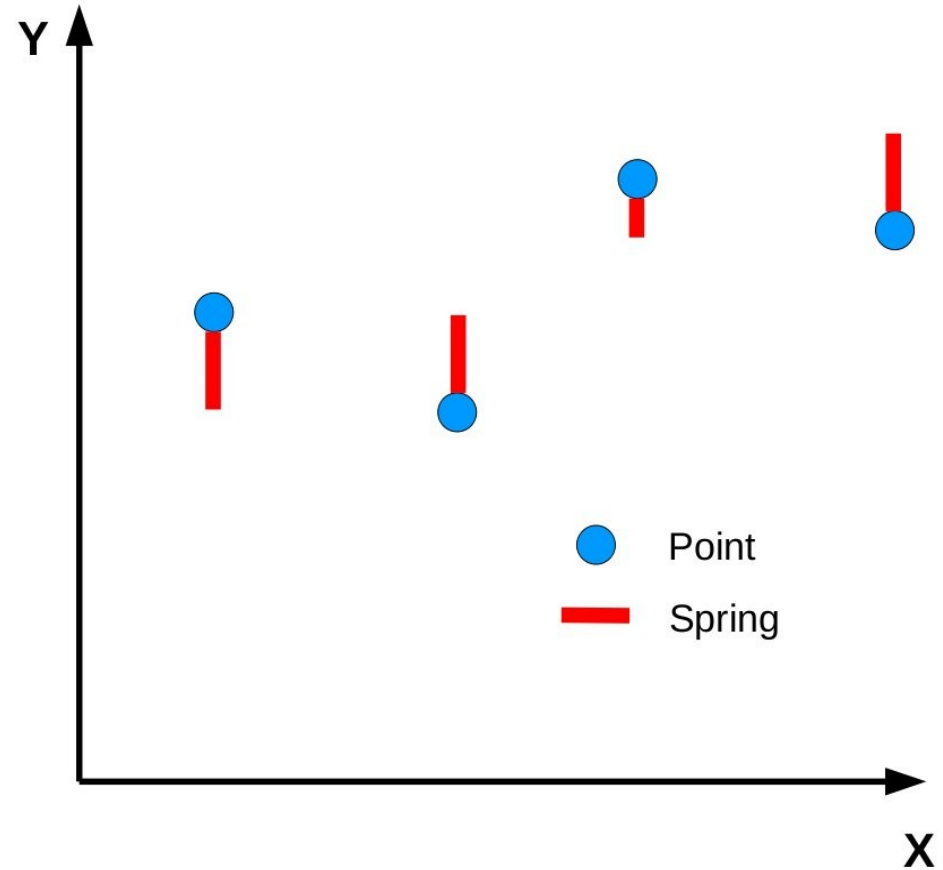
1st: Mechanical Analog

- Suppose we have a set of points (x_i, y_i) .



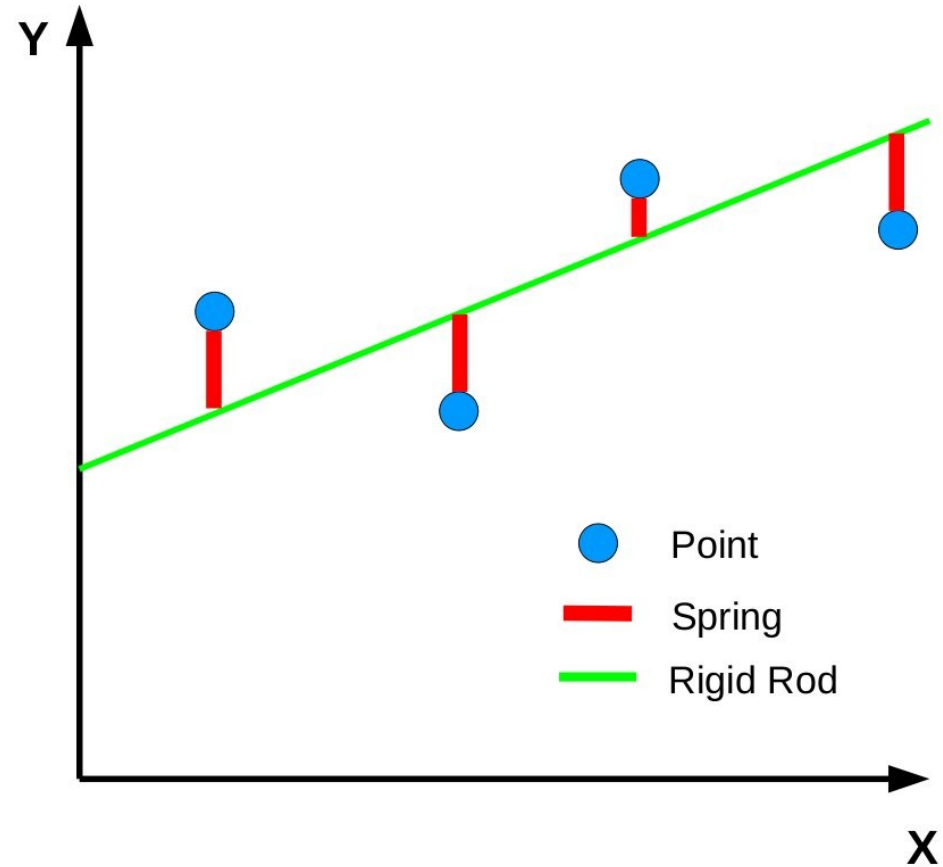
1st: Mechanical Analog

- Suppose we have a set of points (x_i, y_i) .
- To each point, connect a spring. (Imagine spring can only move in y direction.)



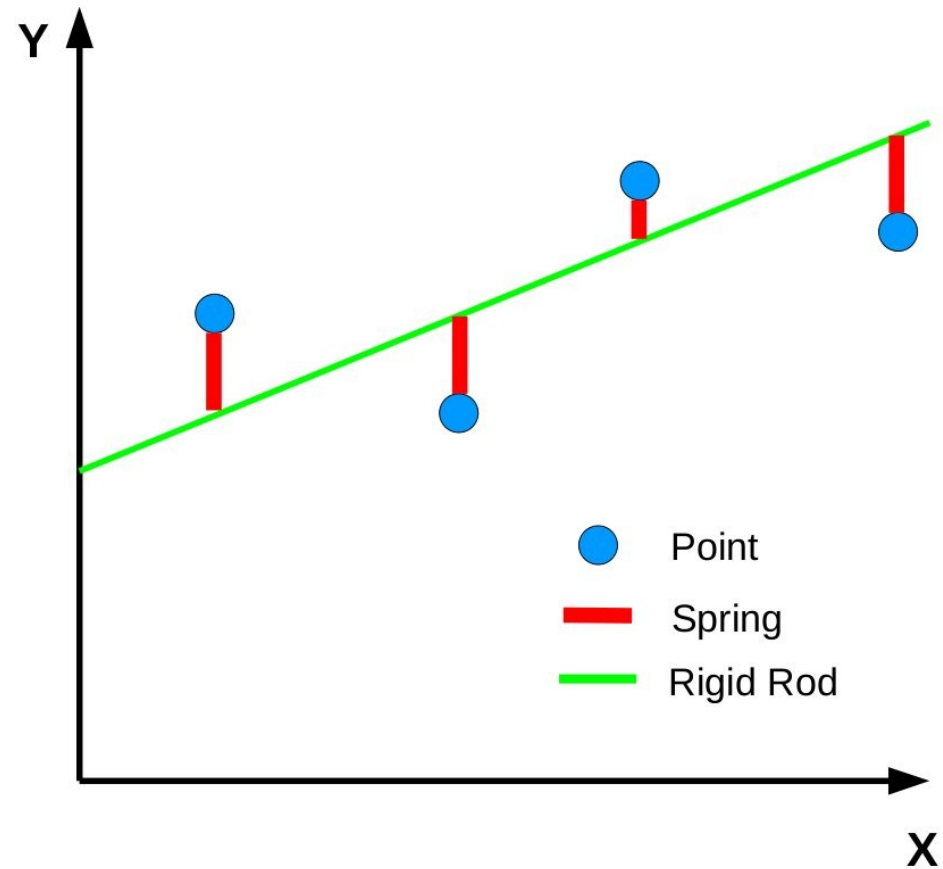
1st: Mechanical Analog

- Suppose we have a set of points (x_i, y_i) .
- To each point, connect a spring. (Imagine spring can only move in y direction.)
- To the other end of the springs attach a rigid rod.



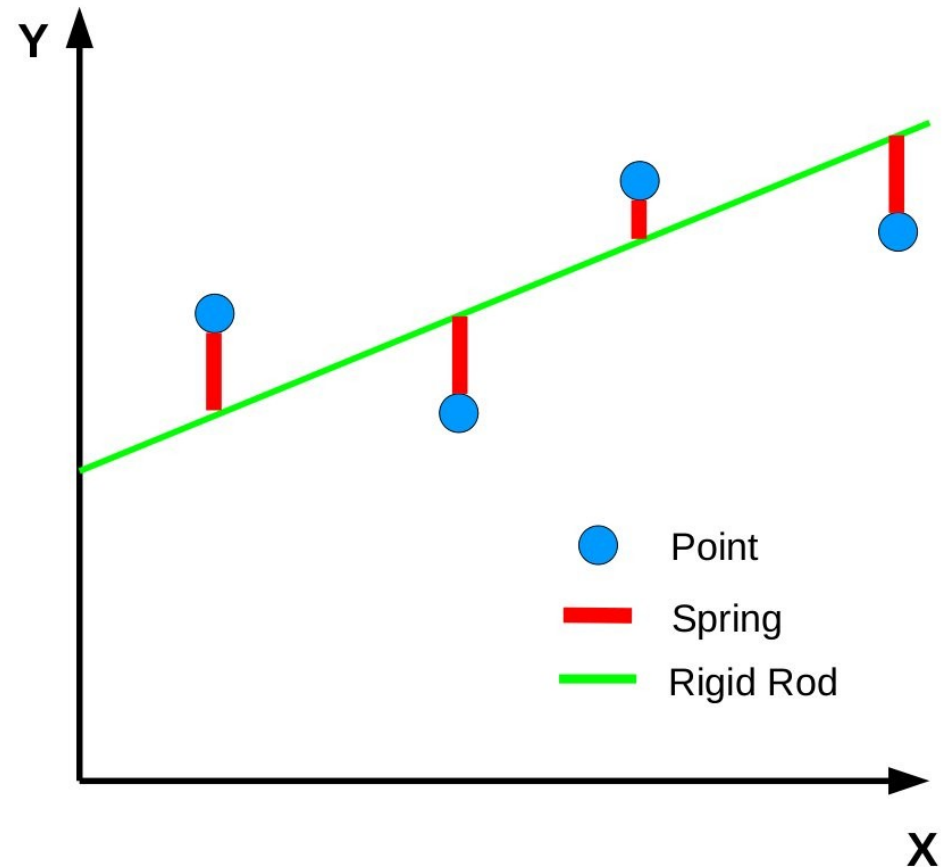
1st: Mechanical Analog

- Suppose we have a set of points (x_i, y_i) .
- To each point, connect a spring. (Imagine spring can only move in y direction.)
- To the other end of the springs attach a rigid rod.
- What will determine where rod comes to rest?



1st: Mechanical Analog

- Suppose we have a set of points (x_i, y_i) .
- To each point, connect a spring. (Imagine spring can only move in y direction.)
- To the other end of the springs attach a rigid rod.
- Equilibrium position of rod will minimize *potential energy*.



replace

$$\frac{1}{2}k_i \longrightarrow \frac{1}{\sigma_i^2}$$

thus replace

$$PE = \sum_{i=1}^N \frac{1}{2}k_i (y'_i - (A + Bx_i))^2$$

**With the
“chi-squared”
function:**

$$\chi^2 = \sum_{i=1}^N \frac{(y'_i - (A + Bx_i))^2}{\sigma_i^2}$$

The problem of fitting data with normally distributed uncertainties is reduced to a matter of finding the model parameters that minimize the chi-squared.

replace

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thus replace

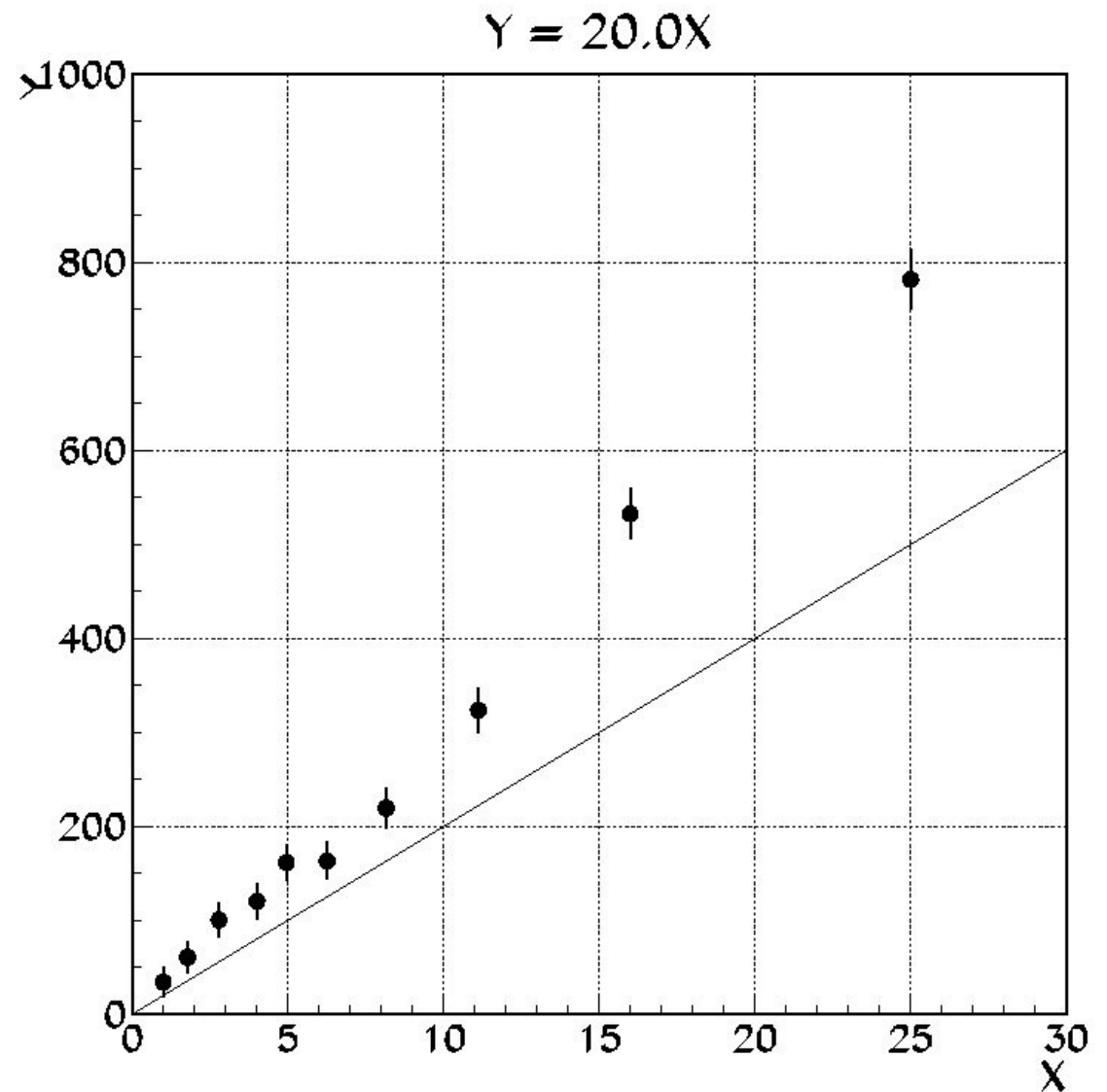
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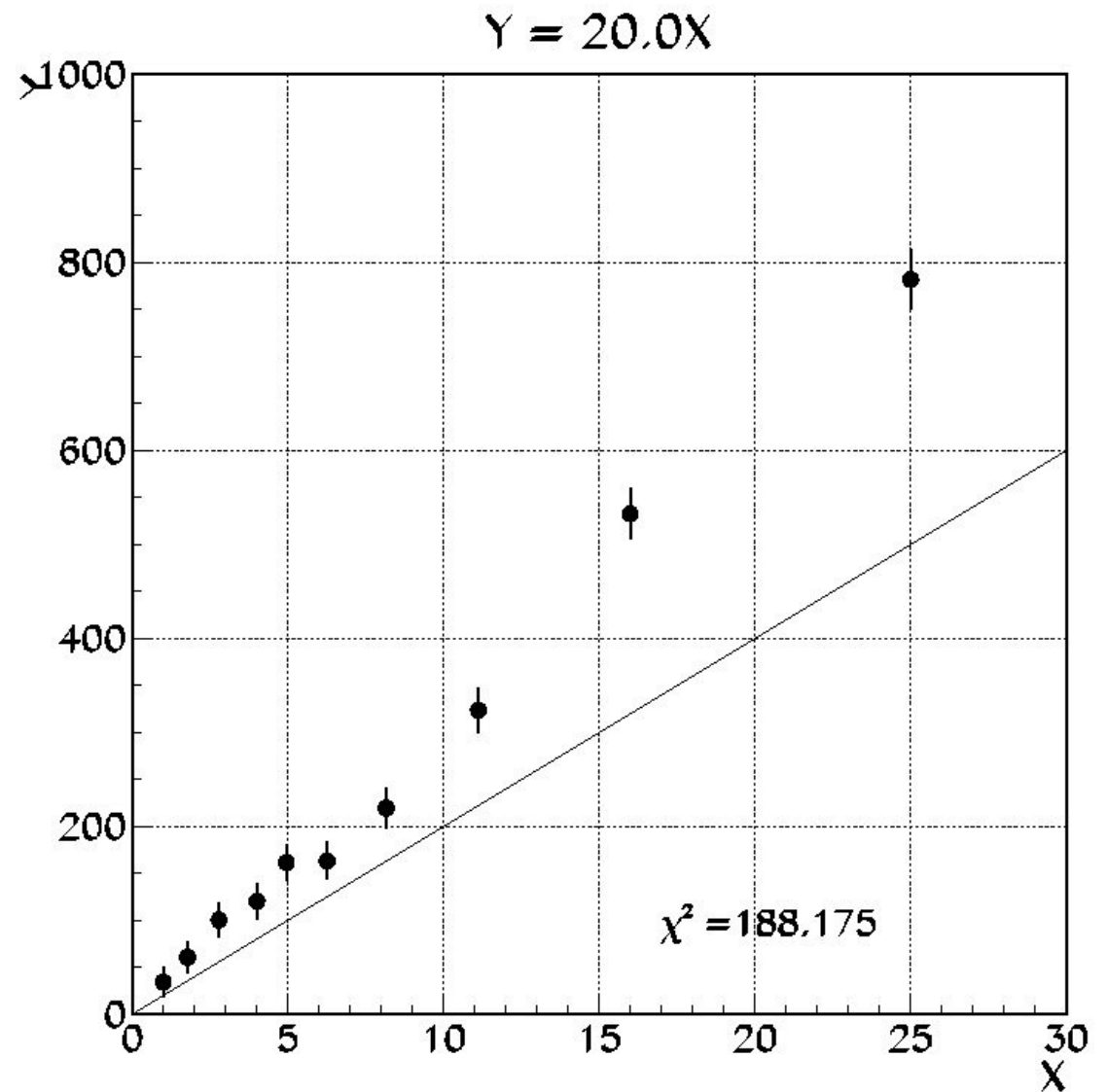
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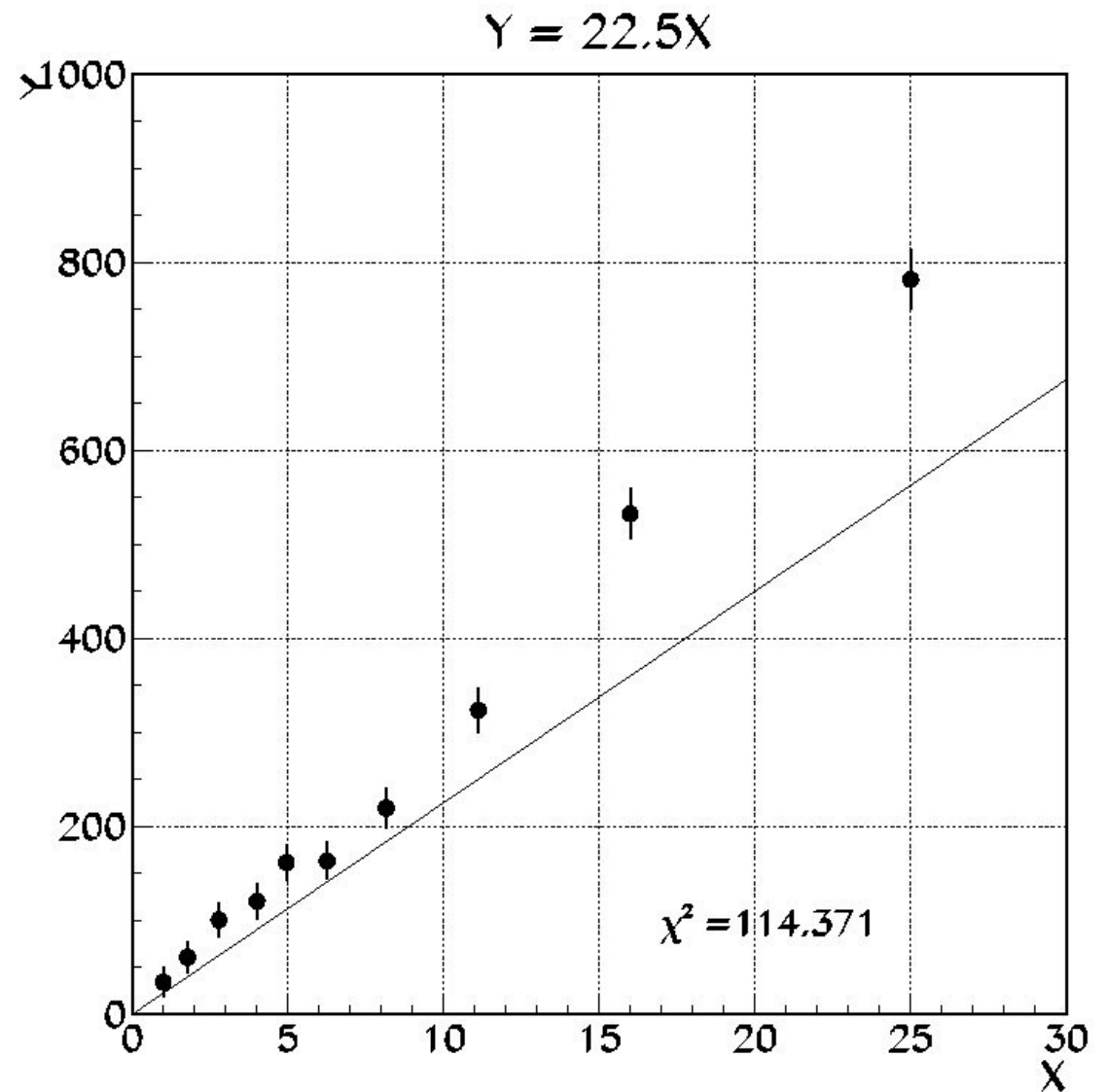
Direct minimization of χ^2 ...



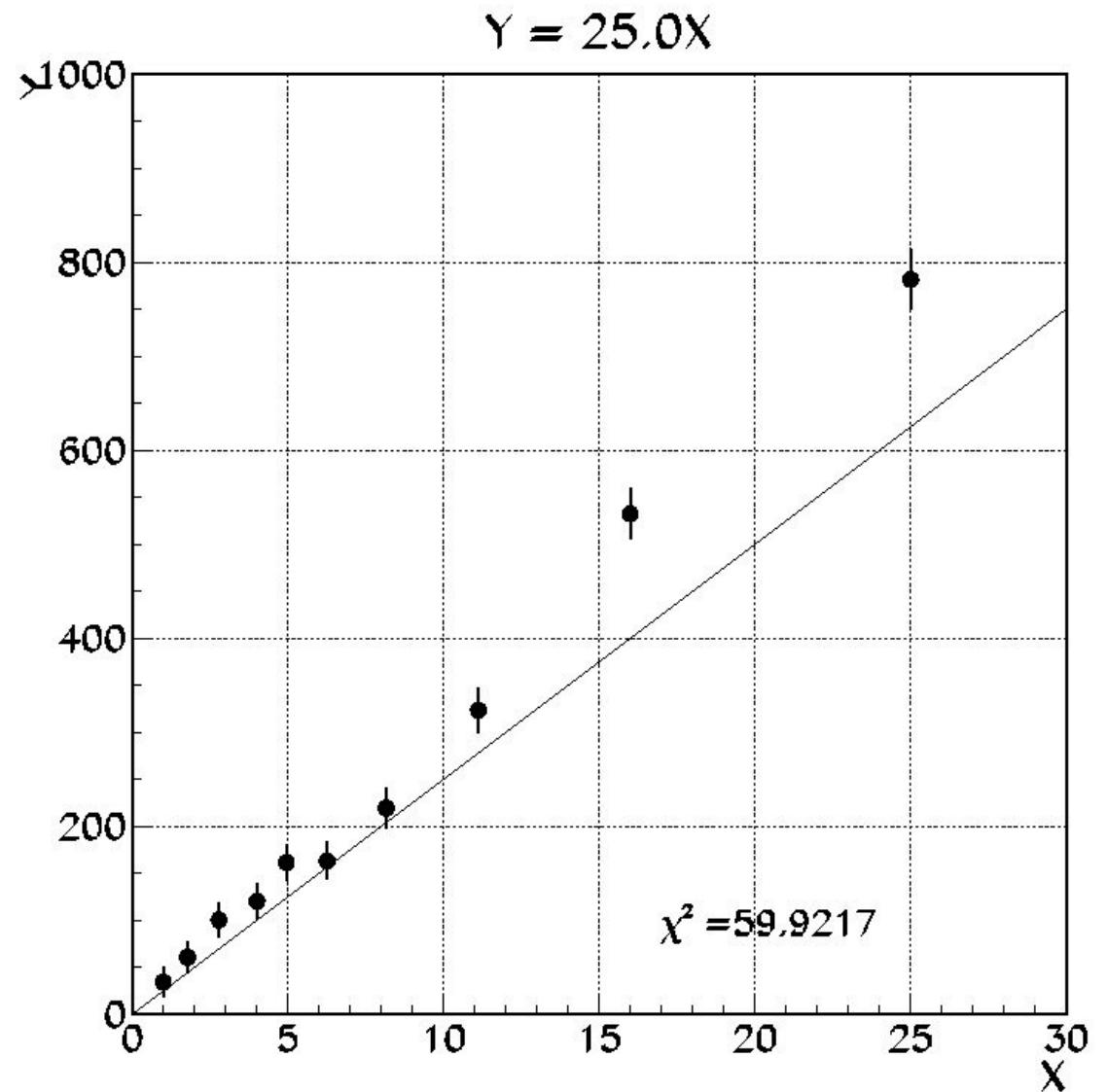
Grid Search minimization of χ^2 ...



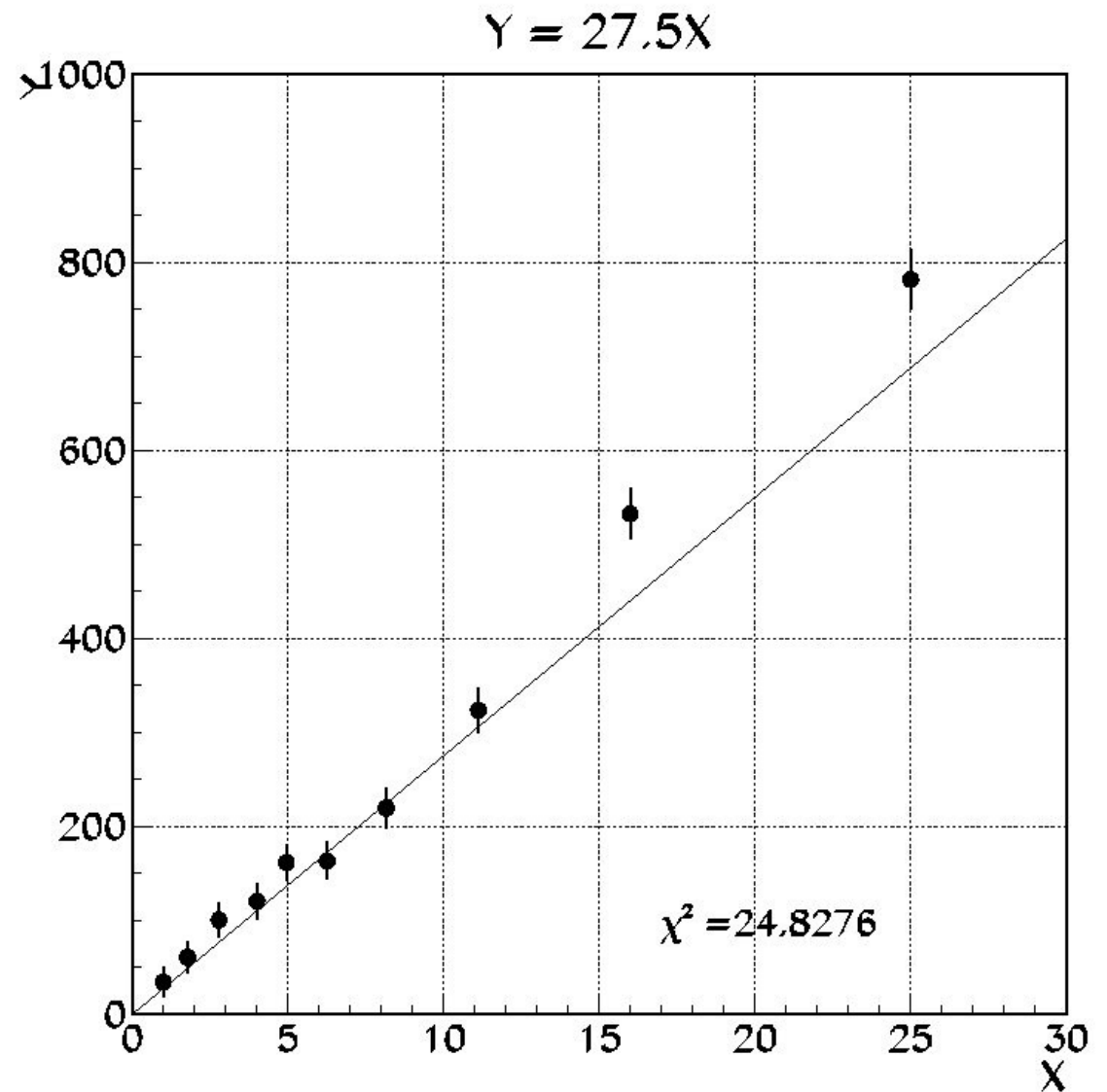
Grid Search minimization of χ^2 ...



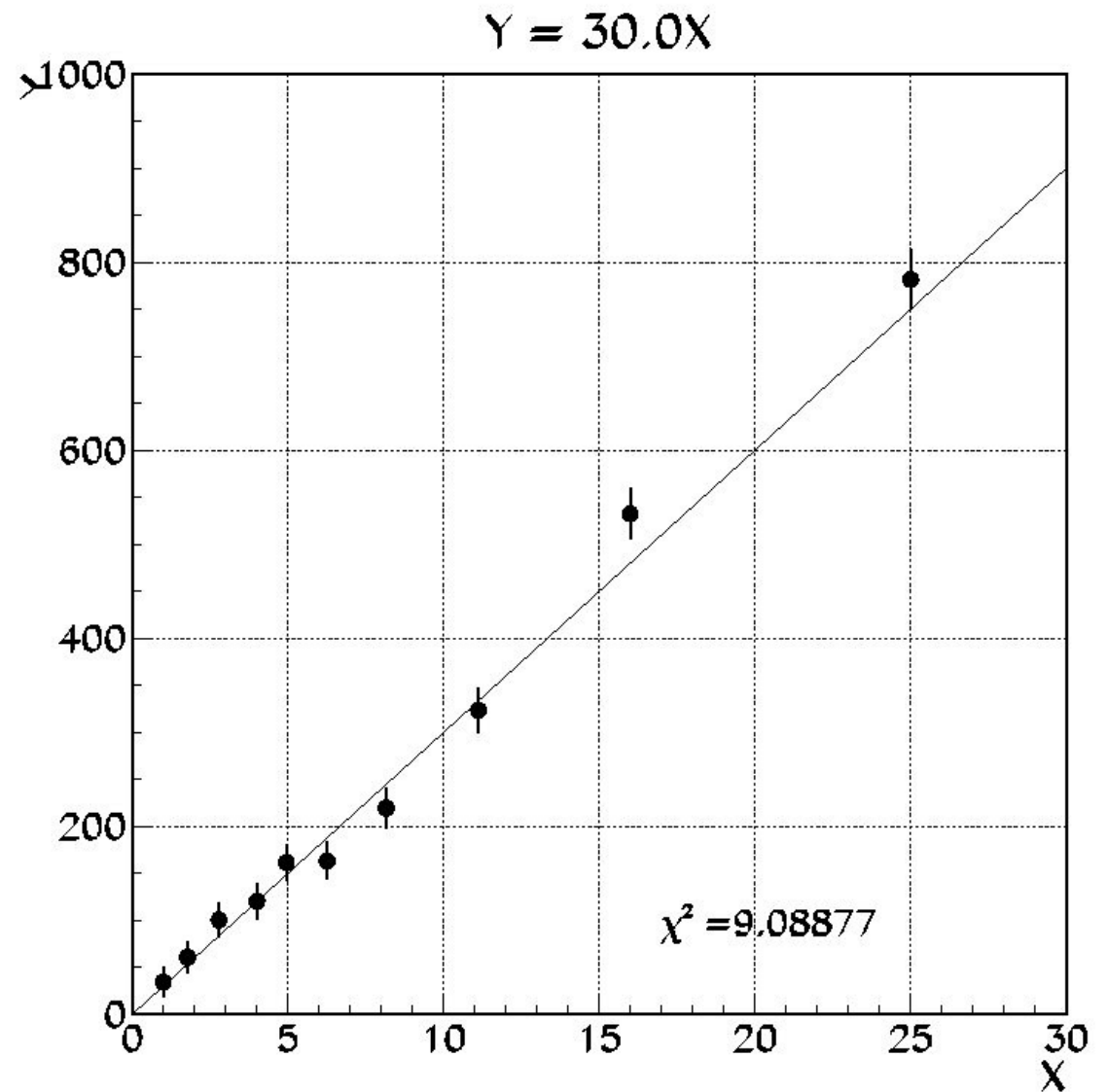
Grid Search minimization of χ^2 ...



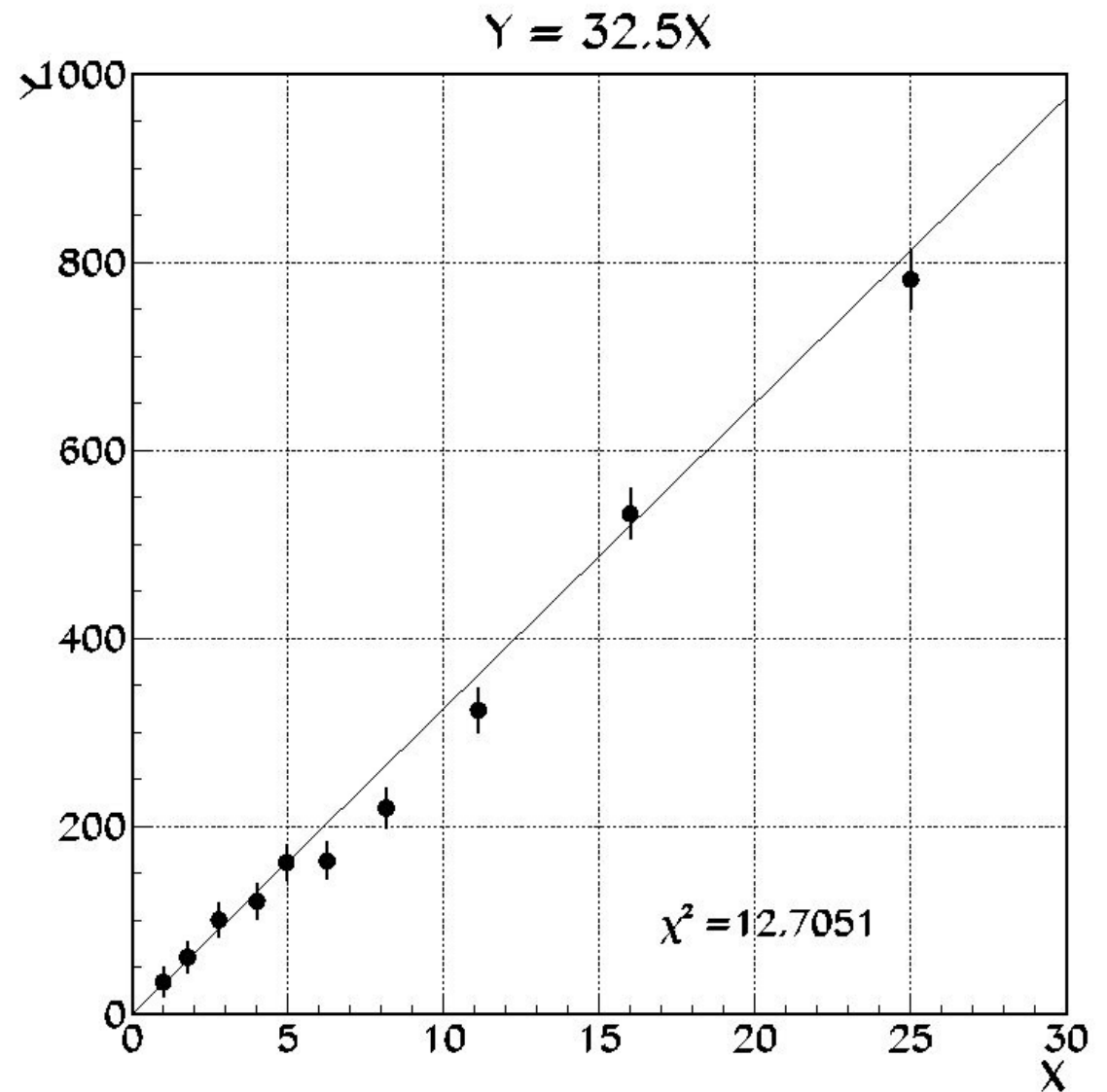
Grid Search minimization of χ^2 ...



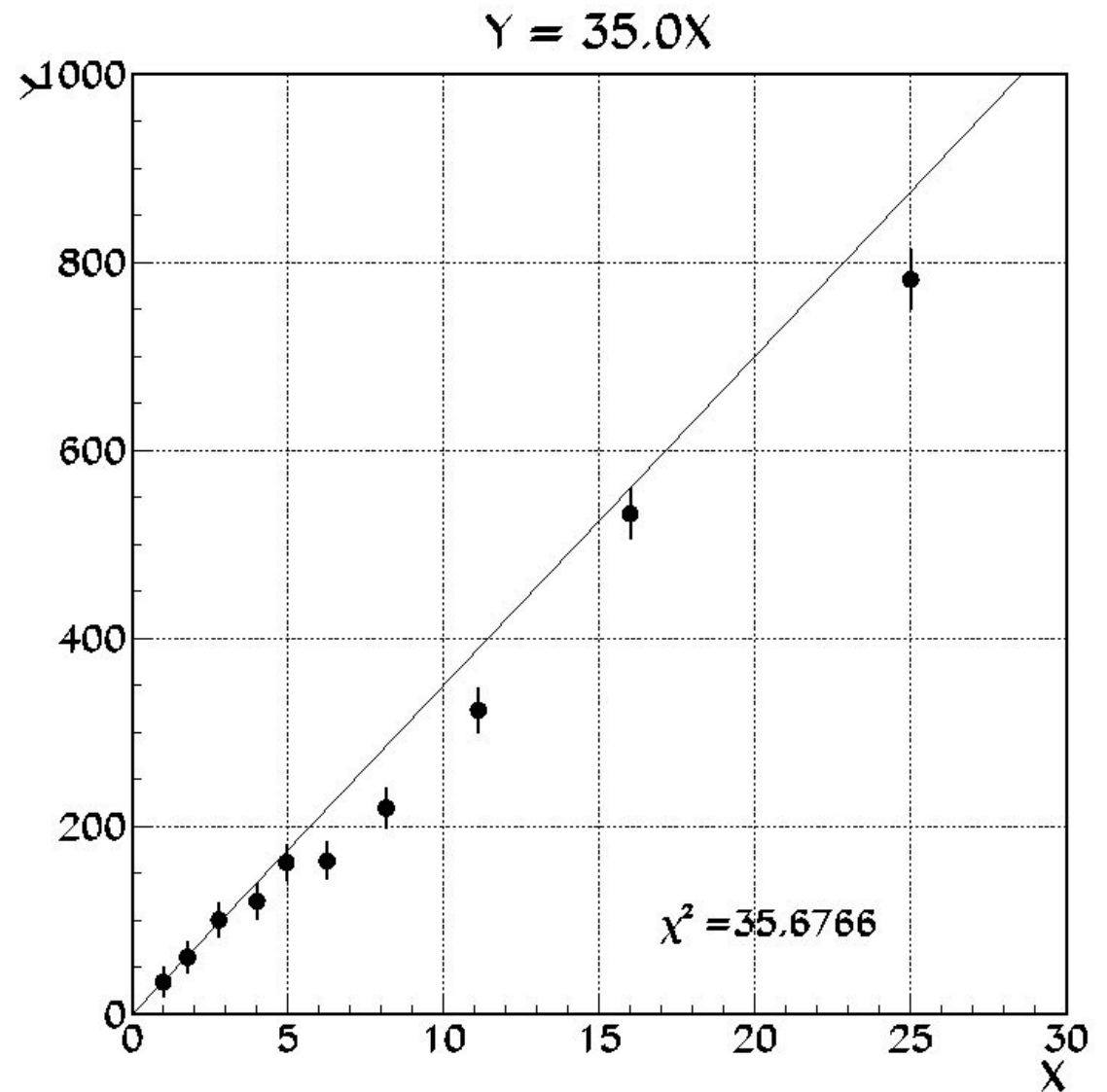
Grid Search minimization of χ^2 ...



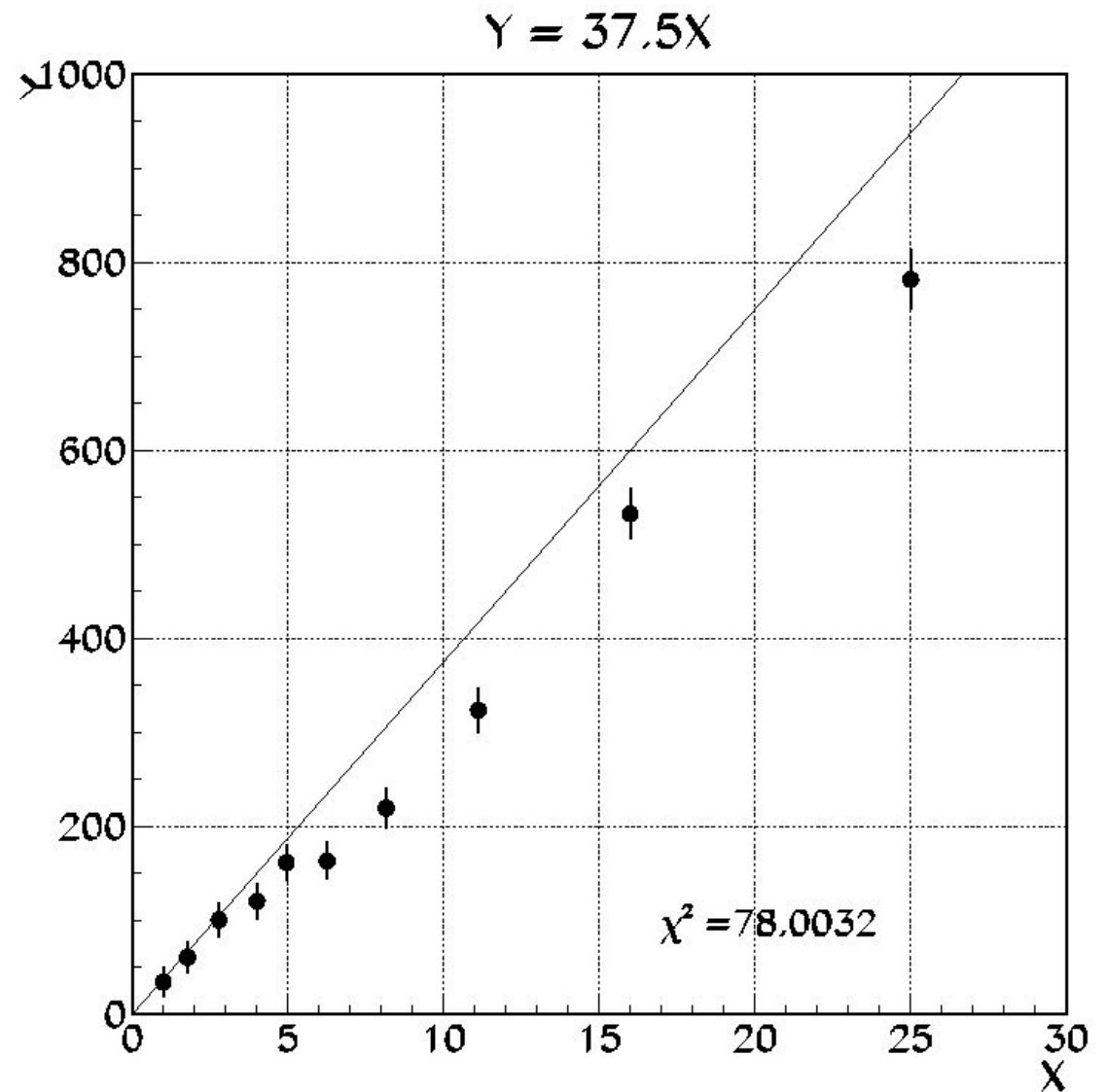
Grid Search minimization of χ^2 ...



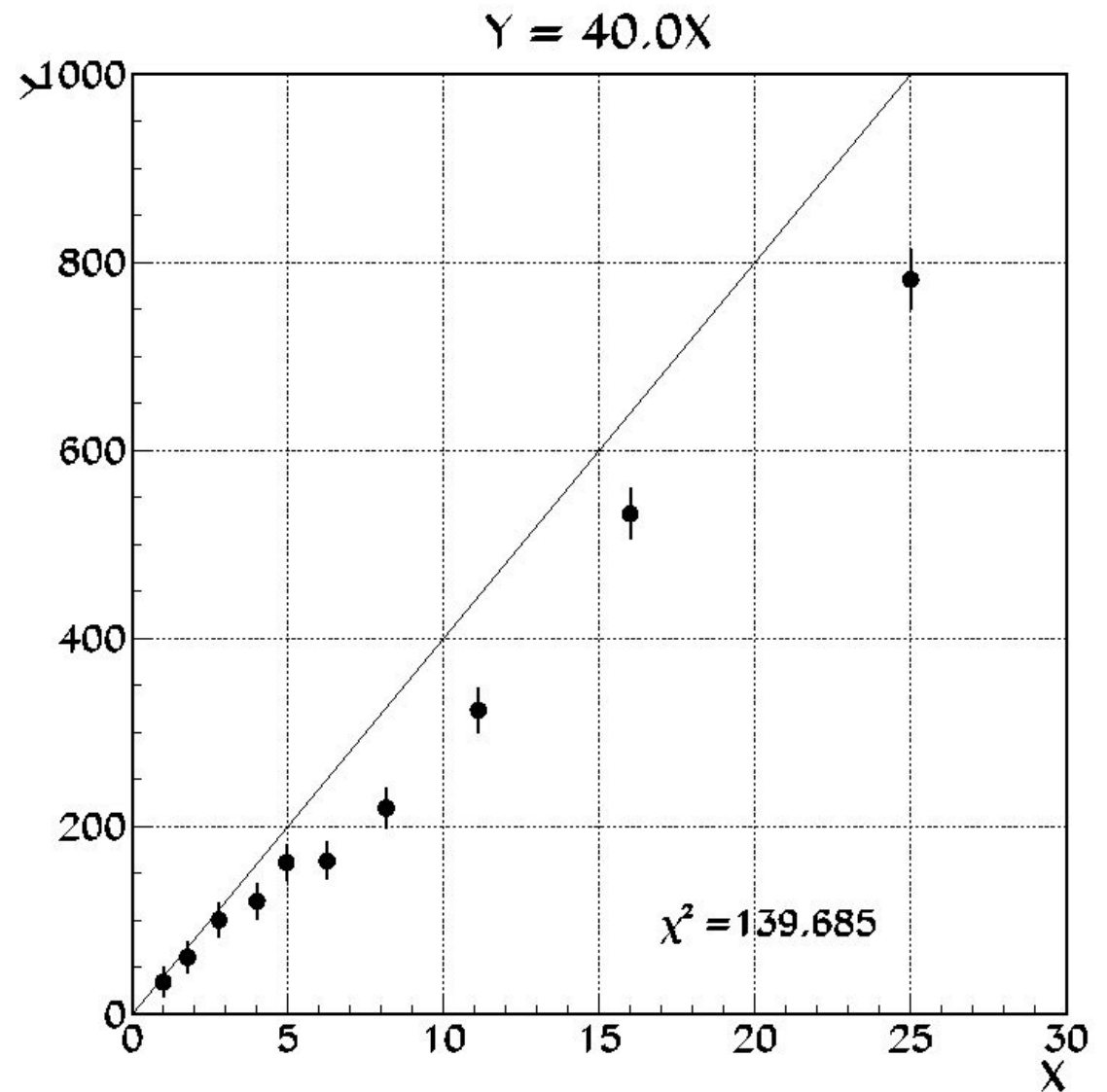
Grid Search minimization of χ^2 ...



Grid Search minimization of χ^2 ...

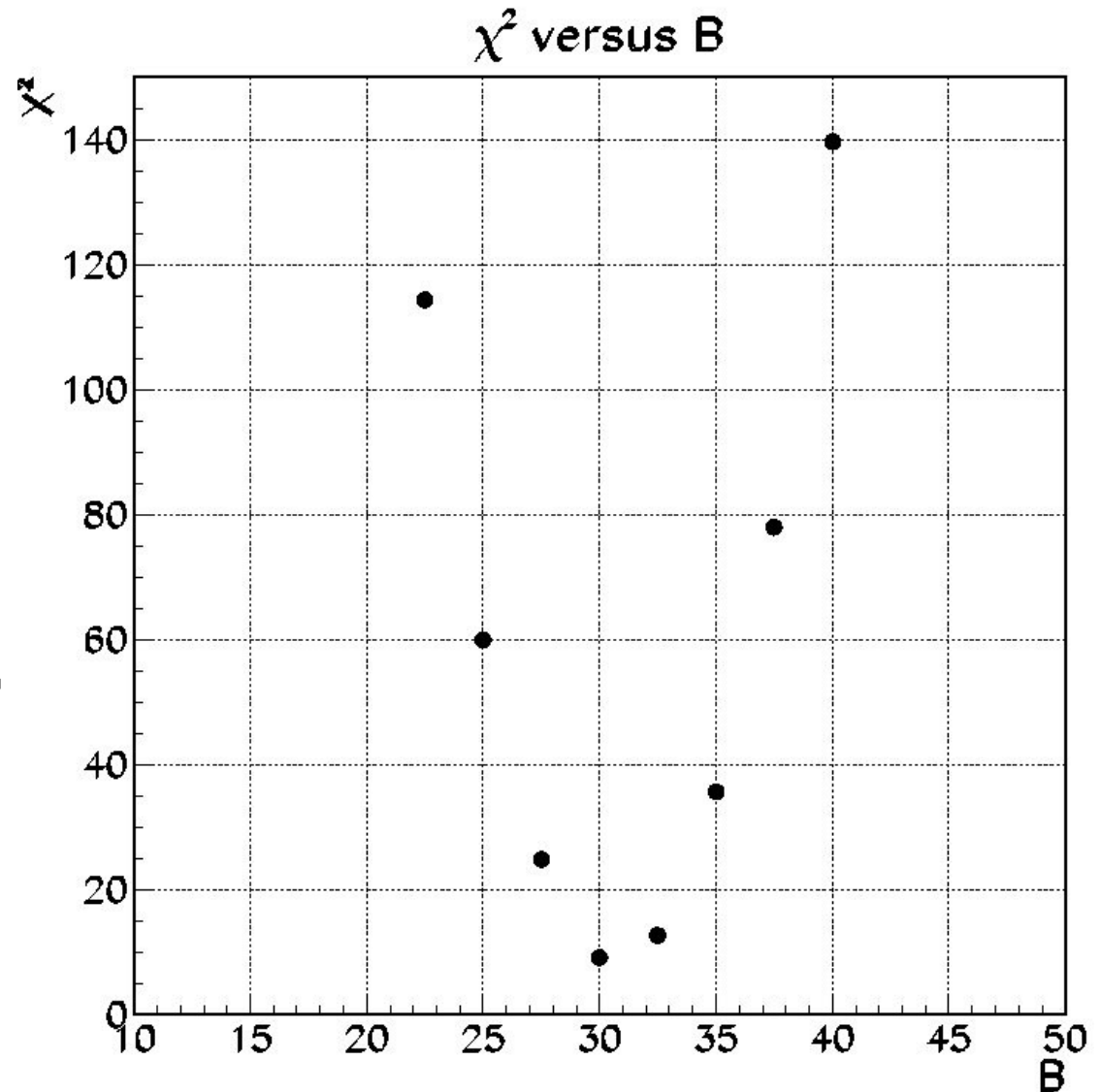


Grid Search minimization of χ^2 ...



Grid Search minimization of χ^2 ...

- “ χ^2 parabola”
- Also can tell us about uncertainty in fit parameters..
- $\chi^2 + 1 \rightarrow B \pm \sigma_B$



Evaluating Fits: χ^2

- Definition of χ^2

$$\chi^2 = \sum_{i=1}^n \frac{(f(x_i) - y_i)^2}{\sigma_{y_i}^2}$$

- Absolute magnitude of χ^2 only has meaning relative to the number of **degrees of freedom**
- **#DOF** = (# of data points) -
(# **parameters determined from the data**)
- **#DOF** = (# of data points) - (# **constraints**)
- $\chi^2 / \text{\#DOF} \sim 1$ for good fit (rule of thumb)
- What would “zero degrees of freedom” mean?

In SciPy...

```
popt, pcov = curve_fit(f,x,y,sigma=sigy)
```

$\text{popt} = (A_{best}, B_{best}, C_{best} \dots)$

**“Optimized
parameters”**

$$\text{pcov} = \begin{pmatrix} \sigma_A^2 & \sigma_{AB} & \sigma_{AC} & \dots \\ \sigma_{AB} & \sigma_B^2 & \sigma_{BC} & \dots \\ \sigma_{AC} & \sigma_{BC} & \sigma_C^2 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}$$

**“Covariance
Matrix”**

Report: $\mathbf{A} = \mathbf{A}_{best} \pm \sigma_A$, et cetera

Are the data and model consistent?

- χ^2 will increase linearly in n
- Absolute magnitude of χ^2 only has meaning relative to the number of **degrees of freedom** in the fit.

$$\chi^2 = \sum_{i=1}^n \frac{(f(x_i) - y_i)^2}{\sigma_{y_i}^2}$$

- **#DOF** = N_{df}
= (**# of data points**) - (**# parameters**)
- $\chi^2 / N_{\text{df}} \sim 1$ for good fit
- But we can do better!

Are the data and model consistent?

True:

1

Goodness-of-fit

False:

0

- “Goodness-of-fit”
 - aka “quality factor”
 - aka “p-value”
 - aka “chance probability”
- Number between 0 → 1
- Quantifies the probability of obtaining your χ^2 / N_{df} by chance.

Are the data and model consistent?

$$Q = \Gamma_{inc}^{up} \left(\frac{N_{df}}{2}, \frac{\chi^2}{2} \right)$$

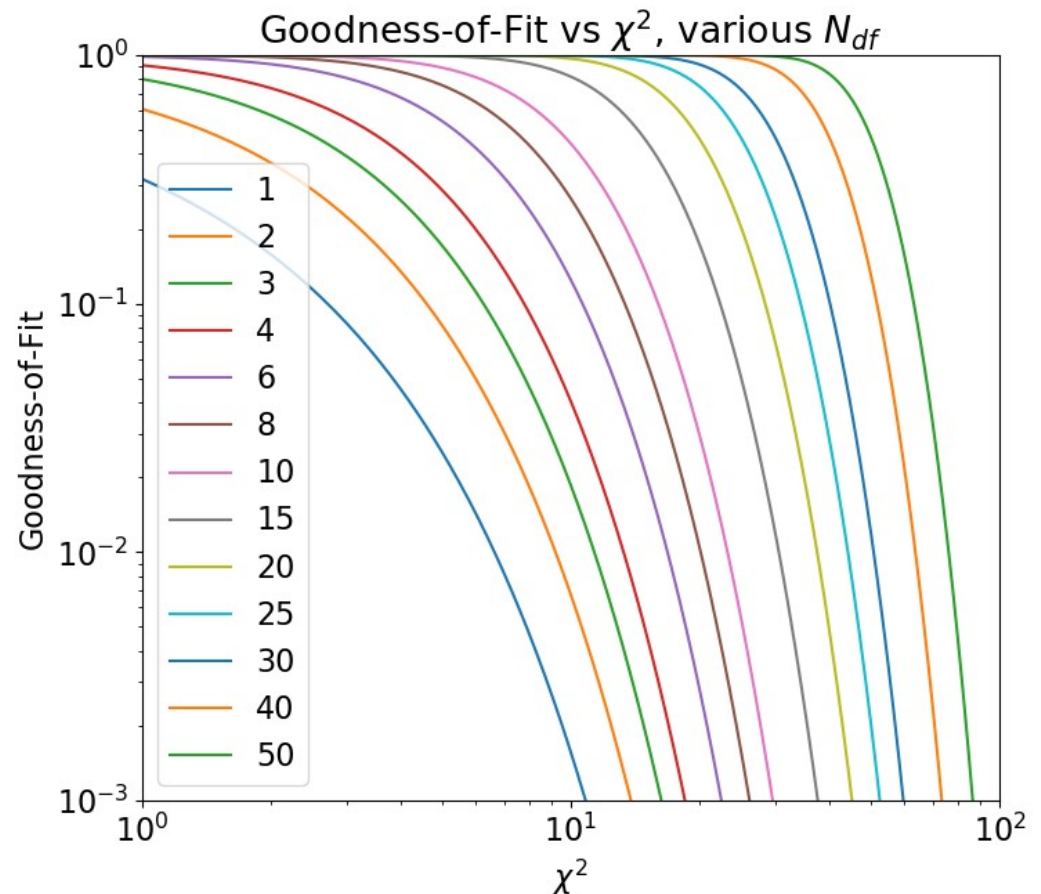
$$\Gamma_{inc}^{up}(a, x) = \frac{1}{\Gamma(a)} \int_x^{\infty} t^{a-1} e^{-t} dt$$

$$\Gamma(a) = \int_0^{\infty} t^{a-1} e^{-t} dt$$

Q = `scipy.special.gammaincc`

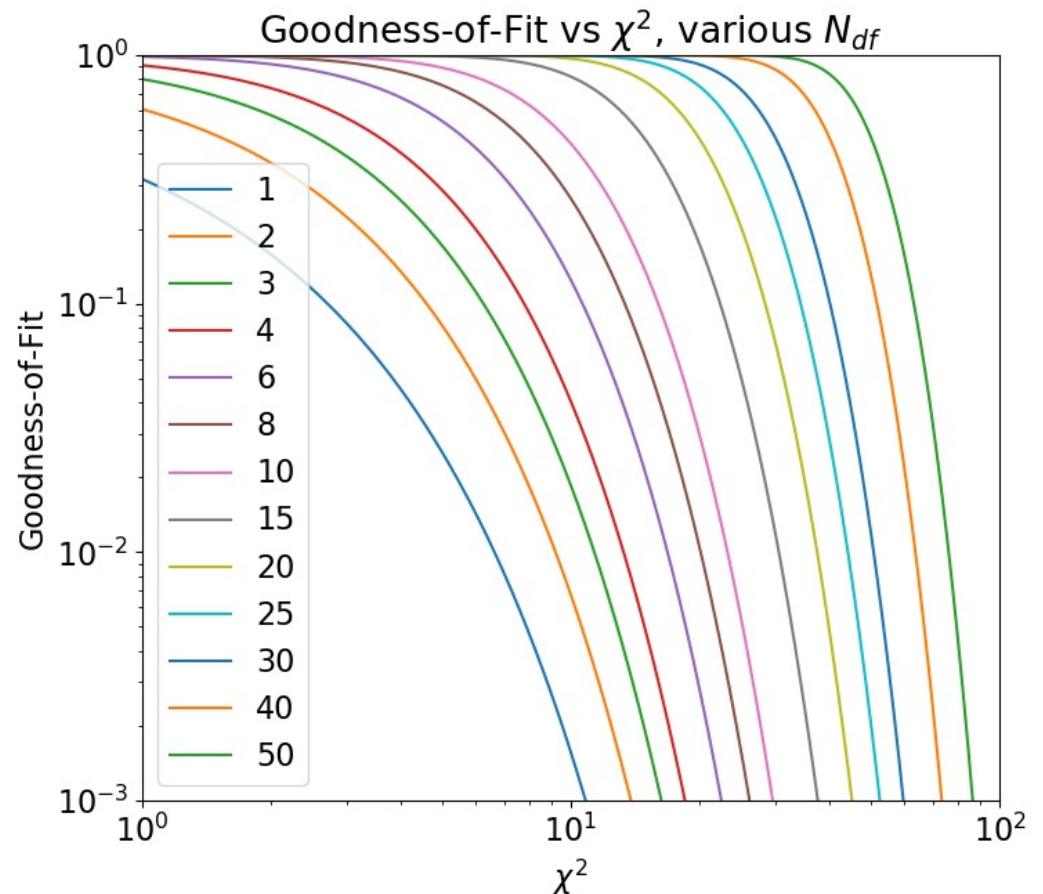
Are the data and model consistent?

- In a fit to experimental data, I obtain
 - $\chi^2 = 10.0$
 - $N_{df} = 6.0$
- Goodness of fit = 0.125
- “If the model were true, I’d expect the same χ^2 or larger in 12.5% of experiments identical to mine.”
- Approximate “two sigma” with $Q = 0.05$ in assignments.



What is the best-pick model?

- It is possible to get a better X^2 by increasing the number of parameters fit.
- If the goodness-of-fit hasn't improved (gotten closer to 1), you haven't gained anything!
- The best model is the one with the largest goodness of fit.



Additional Reading:

Taylor: *An Introduction to Error Analysis*

Bevington: *Data Reduction and Error Analysis for the Physical Sciences*

Press *et al*, *Numerical Recipes*